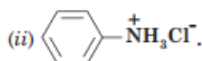
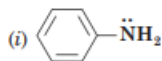




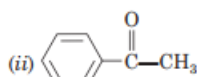
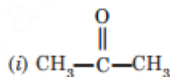
MODULE – III

Short type Questions (2 mark)

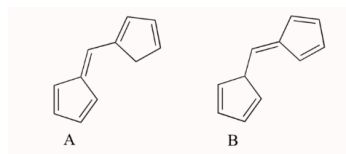
1. Identify the microwave active molecules(s). Give reasons CO, CH₃Cl, H₂O, CH₃-CH₃, CCl₄, CO₂, HCl, NH₃.
2. $\lambda_{\max}$ for aniline shift from 230 nm in neutral medium to 203 nm in acidic medium. Explain
3. The absorbance of a 2.5×10^{-4} M solution taken in a 1 cm is 1.17. Calculate the molar extinction coefficient.
4. What do fundamental and overtone vibrations mean?
5. Explain effect of conjugation on chromophores.
6. What do you mean by chromophore and auxochromes?
7. Explain various type electronic transitions in UV spectroscopy.
8. Find the uncertainty in the position of an electron when the mass of an electron is 9.1×10^{-28} g and the uncertainty in velocity is equal to 2×10^{-3} cm/sec.
9. What are the parameters for expressing absorbance?
10. What is transmittance?
11. Explain the relation between absorbance and transmittance.
12. Calculate the concentration of a substance A in an ethanolic solution of which the absorbance in a 1 cm cell at its $\lambda_{\max} = 241$ nm was found to be 0.890. the molar extinction coefficient of A is 540 at 241 nm.
13. Calculate the concentration in $\mu\text{g ml}^{-1}$ of a solution of substance B (mol wt 204.2) in 1 (M) HCl showing absorbance at its $\lambda_{\max} = 277$ nm of 0.613 in a 4 cm cell. The molar absorptivity at 277 nm is 5432.
14. What are different electronic transitions that take place on absorption of UV light?
15. What are different allowed electronic transitions that take place on absorption of UV light?
16. Mention the application UV spectroscopy.
17. Can UV differentiate ethylbenzene from styrene?
18. Which of the following compound will have greater $\lambda_{\max}$?



19. Which of the following compound will have greater $\lambda_{\max}$?



20. Explain effect of conjugation on absorbance.
21. Differentiate between chromophore and auxochrome.
22. Which of the following molecules would you expect absorb at a longer wavelength in the UV region of the electromagnetic spectrum? Explain your answer.



23. Express $A = 1.0$ in terms of percent transmittance (%T, the unit usually used in IR spectroscopy (and sometimes in UV-vis as well).
24. Give the classification of rigid rotors.
25. Write a note on transitions observed in the rotational spectrum.
26. Define degeneracy and explain it in detail for energy levels $J=1$ and $J=2$
27. Why N_2 molecule is inactive to rotational spectroscopy?
28. Give the equations of fundamental absorption and first overtone.
29. What is mean by zero point energy?
30. Enlist different regions of electromagnetic spectrum.
31. The microwave active molecules are HCl, OCS, CO, CO_2
32. Which of the following transitions are of weak intensities and lie in the visible region?
(a) $n - \pi^*$ (b) $\sigma - \sigma^*$ (c) $\pi - \pi^*$ (d) $n - \sigma^*$
33. What is order of decreasing vibrational frequency for C-Cl, C-Br, C-C, C-O, C-H
34. How many modes of vibrations are there in water molecule? State whether all are IR active?
35. What are the wavelength ranges for UV, IR and Visible regions?
36. Define the terms bathochromic shift and hypochromic shift.
37. What are chromophores? Give some examples.
38. What are auxochromes? Give some examples.
39. What are the Selection rules for Rotational and vibrational spectroscopy?
40. Calculate the fundamental vibrational frequency HCl molecule if the value of force constant of the molecule is 483 Nm^{-1} . The atomic mass $H = 1.6 \times 10^{-27} \text{ kg}$, $Cl = 58 \times 10^{-27} \text{ kg}$
41. Calculate the molar absorptivity of $0.5 \times 10^{-3} \text{ M}$ dye solution in ethanol, which shows an absorbance of 0.17, when 1.3 cm cuvette is used.

Long type Questions (6 marks)

42. The frequency of oscillation of HF is $5 \times 10^{12} \text{ Hz}$. Calculate the force constant and vibrational energy in Joule and cm^{-1} . Calculate the zero-point energy of the molecule.
43. Internuclear distance of HCl molecule (rigid type) is 129 pm. Calculate its rotational constant in cm^{-1} and find the wavelength of the transition between rotational energy levels $J=2$ to $J=3$.
44. The first line in the rotational spectrum of $^{12}\text{C}^{16}\text{O}$ molecule is 3.84235 cm^{-1} . Find out the bond length of the molecule.
45. The Fundamental vibrational frequency of $^1\text{H}^{35}\text{Cl}$ molecule is $86.63 \times 10^{12} \text{ Hz}$. Calculate the zero-point energy and force constant of HCl.
46. Explain rotational constant B.
47. Derive Beer Lamberts law. List its limitations.
48. State Beer-Lambert's law. And derive the expression.
49. What are the various types of electronic transitions in UV-visible spectroscopy?
50. What are the various electronic transitions possible for a molecule?
51. CO molecule absorbs at 2140 cm^{-1} . Calculate the force constant of the molecule, given atomic masses of C and O are 12u and 16 u respectively $1\text{u}: 1.67 \times 10^{-27} \text{ kg}^{-1}$